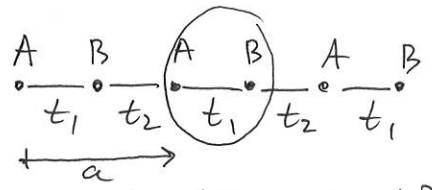
unit cell



$$E\,\phi_n^A = -t_1\,\phi_n^B - t_2\,\phi_{n-1}^B$$

$$E\,\phi_n^B = -t_1\,\phi_n^A - t_2\,\phi_{n+1}^B$$

$$\phi_n^{A/B} = \frac{1}{\sqrt{N}} \sum_k \left(A_k \text{ or } B_k\right) e^{ikna}$$, note a = distance from A to A.

$$E\,A_k = \left(-t_1 - t_2 e^{-ika}\right) B_k$$

$$E\,B_k = \left(-t_1 - t_2 e^{ika}\right) A_k$$

$$\begin{pmatrix} E & t_1 + t_2 e^{-ika} \\ t_1 + t_2 e^{ika} & E \end{pmatrix} \begin{pmatrix} A_k \\ B_k \end{pmatrix} = 0$$

$$E(k) = \pm\sqrt{\left(t_1 + t_2 e^{-ika}\right)\left(t_1 + t_2 e^{ika}\right)} =$$

$$= \pm\sqrt{t_1^2 + t_2^2 + 2t_1 t_2 \cos ka}$$

let now $t_1 = t\,(1+\alpha u)$

$t_2 = t\,(1-\alpha u)$

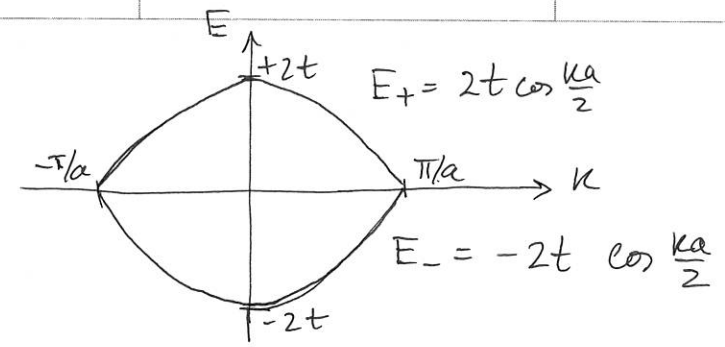
$$E(k) = t\left\{1 + \cancel{2\alpha u} + \alpha^2 u^2 + 1 - \cancel{2\alpha u} + \alpha^2 u^2 + 2\left(1-\alpha^2 u^2\right)\cos ka\right\}^{1/2}$$

$$E(k) = \pm t\sqrt{2}\sqrt{1 + \alpha^2 u^2 + \left(1-\alpha^2 u^2\right)\cos ka}$$

$$\cos ka = \cos^2\frac{ka}{2} - \sin^2\frac{ka}{2} = 1 - 2\sin^2\frac{ka}{2} = 2\cos^2\frac{ka}{2} - 1$$

$$E_\pm(k) = \pm 2t\sqrt{\cos^2\frac{ka}{2} + \alpha^2 u^2 \sin^2\frac{ka}{2}}$$

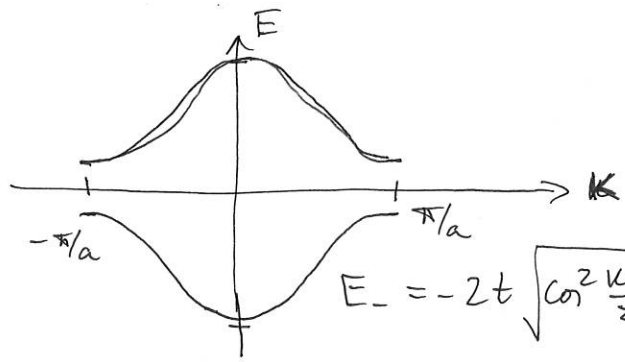
$u=0$



$E_+ = 2t\cos\frac{ka}{2}$

$E_- = -2t\cos\frac{ka}{2}$

$u\neq 0$



$E_- = -2t\sqrt{\cos^2\frac{ka}{2} + \alpha^2 u^2 \sin^2\frac{ka}{2}}$

Expand near $\frac{\pi}{a} \Rightarrow k = \frac{\pi}{a} + q$ and expand is small q:

$$\cos^2\frac{ka}{2} = \cos^2\left(\frac{\pi}{2} + \frac{qa}{2}\right) = \sin^2\left(\frac{qa}{2}\right) \approx \left(\frac{qa}{2}\right)^2$$

$$\sin^2\frac{ka}{2} = \sin^2\left(\frac{\pi}{2} + \frac{qa}{2}\right) = \cos^2\left(\frac{qa}{2}\right) \approx 1 - \left(\frac{qa}{2}\right)^2$$

$$E_-(q) \cong -2t\sqrt{\frac{q^2a^2}{4} + \alpha^2u^2\left(1 - \frac{q^2a^2}{4}\right)} =$$

$$= -t\sqrt{\alpha^2u^2 + (1-\alpha^2u^2)q^2a^2} = -t\sqrt{\alpha^2u^2 + q^2a^2}$$

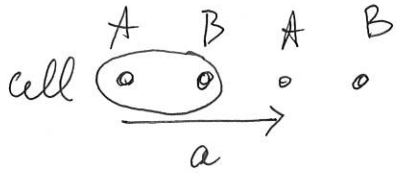
since $(\alpha u)^2 \cdot (qa)^2$ is higher order smallness

while for $u=0$ we get $E_-^{(u=0)}(q) = -ta|q|$ in this approx.

The energy difference between the ~~dimer~~ non-dimerized $(u=0)$ and dimerized $(u\neq 0)$ cases is $(L = Na)$

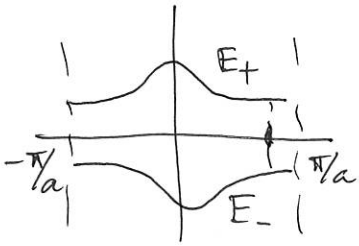
$$\Delta E = 2L\int_{-\pi/a}^{\pi/a}\frac{dk}{2\pi}\left\{E_-^{(u=0)}(k) - E_-(k)\right\}$$

$$= 4L\int_0^{\pi/a}\frac{dk}{2\pi}\left\{E_-^{(u=0)} - E_-\right\}$$



1) Number of k-states in 1BZ: $\dfrac{\frac{2\pi}{a}}{\frac{2\pi}{L}} = \dfrac{L}{a} = N_{cell}$

Each can ~~hold~~ host 2 electrons (spin ↑ and ↓)



Two bands ⇒ total number of electrons to occupy is $2 \cdot 2 \cdot N_{cell}$

2) Each atom, A and B, give 1 electron
⇒ $2 \cdot N_{cell}$ of them in 1BZ

⇒ E_- is completely filled
E_+ is empty.

Now use our approximate form for $E_-(q)$:

$$\Delta E = \frac{4L}{2\pi}\int_{-\pi/a}^{0} dq\left(-ta|q| + t\sqrt{\alpha^2u^2+q^2a^2}\right) \qquad x = qa$$

$$= \frac{4tL}{2\pi a}\int_{-\pi}^{0} dx\left(\sqrt{\alpha^2u^2+x^2} - |x|\right)$$

We now split $\int_{-\pi}^{0} dx \ldots \approx \underbrace{\int_{-\alpha u}^{0} dx \ldots}_{I_1} + \underbrace{\int_{-\pi}^{-\alpha u} dx \ldots}_{I_2}$

In I_1, we can expand in small x:

$$\sqrt{\alpha^2u^2+x^2} \approx \alpha u\left(1+\frac{x^2}{2\alpha^2u^2}\right) = \alpha u + \frac{x^2}{2\alpha u}$$

In I_2, we expand in small αu instead:

$$\sqrt{\alpha^2u^2+x^2} \approx |x|\left(1+\frac{\alpha^2u^2}{2x^2}\right) = |x| + \frac{\alpha^2u^2}{2|x|}$$

(Remember that $\sqrt{x^2} = |x|$!)

$$\Delta E = \frac{2tL}{\pi a}\left\{\int_{-\alpha u}^{0} dx\left(\alpha u + \frac{x^2}{2\alpha u} + x\right)\right.$$

$$\left. + \int_{-\pi}^{-\alpha u} dx\left(|x| + \frac{\alpha^2u^2}{2|x|} - |x|\right)\right\}$$

Therefore $I_1 \sim (\alpha u)^2$ while

$$I_2 = \frac{1}{2}\alpha^2u^2\int_{-\pi}^{-\alpha u}\frac{dx}{-x} = -\frac{1}{2}\alpha^2u^2\int_{\pi}^{\alpha u}\frac{dy}{y}$$

$(x=-y)$

$$I_2 = -\frac{1}{2}\alpha^2 u^2 \ln\left(\frac{\alpha u}{\pi}\right) = \frac{1}{2}\alpha^2 u^2 \ln\left(\frac{\pi}{\alpha u}\right)$$

For small αu $I_2 \gg I_1$ ~~because~~ due to large logarithm $\ln\left(\frac{\pi}{\alpha u}\right)$.

In that factor π under log is really our approximation, it is better to write $\ln\left(\frac{\pi}{\alpha u}\right) \to \ln\left(\frac{\Lambda}{\alpha u}\right)$

where Λ is some numerical factor which we do not know (but which can be calculated).

With this approximation

$$\Delta E = \frac{tL}{\pi a}\alpha^2 u^2 \ln\left(\frac{\Lambda}{\alpha u}\right).$$

$$L = Na \Rightarrow \Delta E = N\,\frac{t\alpha^2 u^2}{\pi}\ln\left(\frac{\Lambda}{\alpha u}\right)$$

Elastic energy:

Start with $\mathcal{H} = \sum_n \frac{p^2}{2m} + \frac{1}{2} k (x_n - x_{n+1})^2$

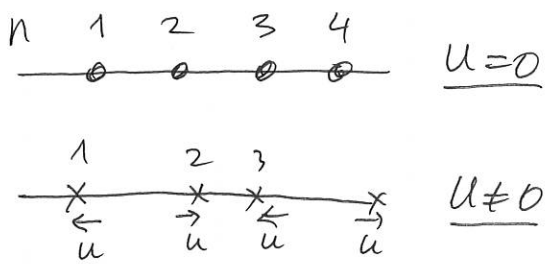
for the <u>non-dimerized</u> chain.

The ground state energy of the non-dimerized chain is therefore

$$E_{elastic}(u=0) = 0.$$

Now consider <u>dimerization</u> by parameterizing

$$x_n = (-1)^n u$$



$$\Rightarrow E_{elastic}(u \neq 0) = \sum_n \frac{1}{2} k \left((-1)^n u - (-1)^{n+1} u\right)^2$$

$$= \sum_n \frac{1}{2} k u^2 \left(1 - (-1)\right)^2 =$$

$$= 2 k u^2 \sum_n (1) = 2 k u^2.$$

<u>Elastic</u> energy of distorted (dimerized) chain is naturally greater than that of non-dimerized chain.

The total energy balance

$$E_{TOT}(u=0) = E_{elastic}(u=0) + E_{electron}(u=0)$$

$$E_{TOT}(u\neq 0) = E_{elastic}(u=0) + E_{electron}(u\neq 0)$$

$$\Rightarrow \Delta\mathcal{E} = E_{TOT}(u=0) - E_{TOT}(u\neq 0) = -2ku^2N + \Delta E$$

$$= -2ku^2N + N\frac{\alpha^2u^2t}{\pi}\ln\left(\frac{\Lambda}{\alpha u}\right)$$

Optimize $\Delta\mathcal{E}$ over u:

$$\frac{1}{N}\frac{\partial(\Delta\mathcal{E})}{\partial u} = 0 = -4ku + \frac{2\alpha^2tu}{\pi}\ln\left(\frac{\Lambda}{\alpha u}\right) + \frac{\alpha^2u^2t}{\pi}\frac{1}{u}$$

$$\Rightarrow u\left(\frac{2\alpha^2t}{\pi}\ln\frac{\Lambda}{\alpha u} + \frac{\alpha^2t}{\pi} - 4k\right) = 0$$

$u=0$ gives $\Delta\mathcal{E}=0$, while the other solution

$$\frac{2\alpha^2t}{\pi}\ln\frac{\Lambda}{\alpha u} \approx 4k - \frac{\alpha^2t}{\pi}$$

$$\ln\frac{\Lambda}{\alpha u} = \frac{2\pi k}{\alpha^2t} - \frac{1}{2}$$

$$\frac{\Lambda}{\alpha u} = e^{+\frac{2\pi k}{\alpha^2t} - \frac{1}{2}} \qquad \alpha u = \Lambda e^{-\frac{2\pi k}{\alpha^2t} + \frac{1}{2}}$$

Therefore

$$\Delta \mathcal{E} = N \left\{ \frac{t}{\pi} \alpha^2 u^2 \left(\frac{2\pi k}{\alpha^2 t} - \frac{1}{2} \right) - \frac{2k}{\alpha^2} \alpha^2 u^2 \right\}$$

$$= N \left\{ 2k u^2 - \frac{1}{2} \frac{\alpha^2 u^2 t}{\pi} - 2k u^2 \right\}$$

$$= - \frac{\alpha^2 u^2 t}{2\pi} N$$

The dimerized state gains energy!